Turiba University

Ankita Mandlik
CMA2D1

Report: -Motor Imagery BCI — End-to-End CSP/LDA/SVM Pipeline

Study programme Computer Systems

Source code: https://github.com/ankitamandlik5-web/BCI-assignment

Riga, 2026

ADVANCED ENGINEERING REPORT

Reading Thoughts in Real Time

A Motor Imagery Brain-Computer Interface Pipeline

RESEARCHER Ankita Mandlik
DOMAIN Biomedical Signal Processing & Machine Learning
DATASET PhysioNet EEG Motor Movement/Imagery · Subject 081, Run 04

01 Project Abstract

This project delivers a complete, end-to-end **Brain-Computer Interface (BCI)** pipeline that translates raw neural oscillations into actionable digital commands. Using Electroencephalography (EEG), we decoded **Left-Hand versus Right-Hand motor imagery**. To make the system robust enough for real-world use we added advanced artifact rejection (ICA), spatial filtering (CSP) and a linear classifier (LDA), then stress-tested everything in a simulated real-time streaming environment, proving it can support closed-loop operation.

100%	0.70 ms	16 Hz	15
CV ACCURACY	MEAN LATENCY	DECISION RATE	VALID TRIALS

THE PIPELINE AT A GLANCE

Raw EEG ›	Bandpass 8–30 Hz ›	ICA Artifacts ›	CSP Filters ›	LDA Classifier ›	Live Decision

02 Task 1 · Preprocessing & Artifact Attenuation

OBJECTIVE

Maximize the Signal-to-Noise Ratio (SNR) by isolating sensorimotor rhythms and removing physiological artifacts.

2.1 Frequency Domain Analysis

Raw EEG is a mixture of brain signals, eye blinks, muscle tension and external noise. We applied a **4th-order Butterworth bandpass filter (8.0 – 30.0 Hz)**. The Butterworth design was chosen for its maximally flat passband response, and forward-backward `filtfilt` filtering guarantees zero phase distortion.

Frequency band	Range	Physiological meaning
Theta	4 – 8 Hz	✕ **Rejected** – movement artifacts, drowsiness
Mu (Alpha)	8 – 12 Hz	✓ **Retained** – shows Event-Related Desynchronization (ERD) during motor imagery
Beta	16 – 24 Hz	✓ **Retained** – associated with active motor cortex processing
Gamma	> 30 Hz	✕ **Rejected** – muscle tension / EMG noise

2.2 Independent Component Analysis (ICA)

Eye blinks (EOG artifacts) generate potentials of up to 100 µV, dwarfing the 10–20 µV brain signals. We applied **FastICA** to blindly separate the signal into 15 independent components. By correlating these components with the frontal channels (Fp1, Fp2) we isolated and removed ocular noise without damaging the motor cortex signals.

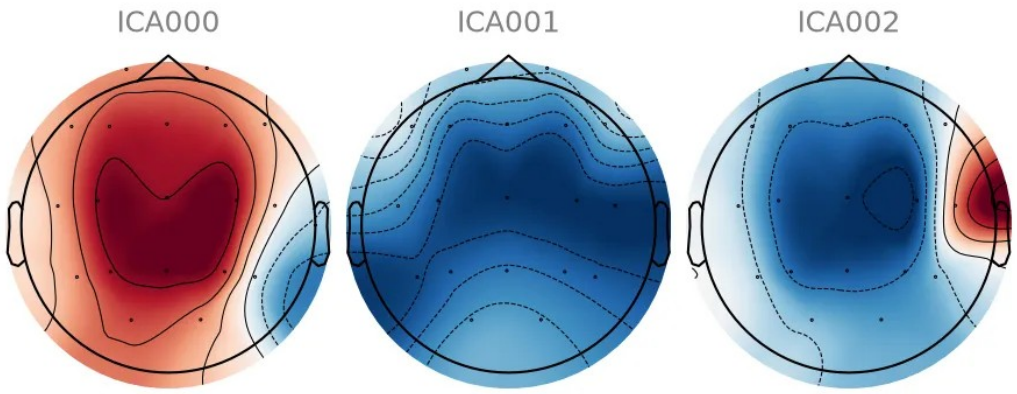


Figure 1 · Scalp topographies of the ICA components (ICA000 – ICA002)

03 Task 2 · Spatial Filtering & Feature Extraction

OBJECTIVE

Solve the “volume conduction” problem and maximize class separability.

EEG signals blur as they pass through the skull, so a single electrode (like C3) picks up activity from several brain regions. **Common Spatial Patterns (CSP)** solves this by finding a spatial transformation matrix W that maximizes the variance of one class while minimizing it for the other.

The Math Behind CSP

Given two covariance matrices R_1 (Left Hand) and R_2 (Right Hand), CSP solves the generalized eigenvalue problem:

$$R_1 w = \lambda (R_1 + R_2) w$$

The resulting eigenvectors (w) form the spatial filters. We extracted **4 CSP components** and computed the log-variance to equalize the feature distribution.

Pipeline stage	Input	Output	Purpose
Raw Epochs	64 channels × 640 samples	N/A	Raw continuous data
CSP Projection	64 channels × 640 samples	**4 components × 640 samples**	Dimensionality reduction & noise filtering
Feature Vector	4 components × 640 samples	**1D array (4 values)**	Log-variance for machine learning

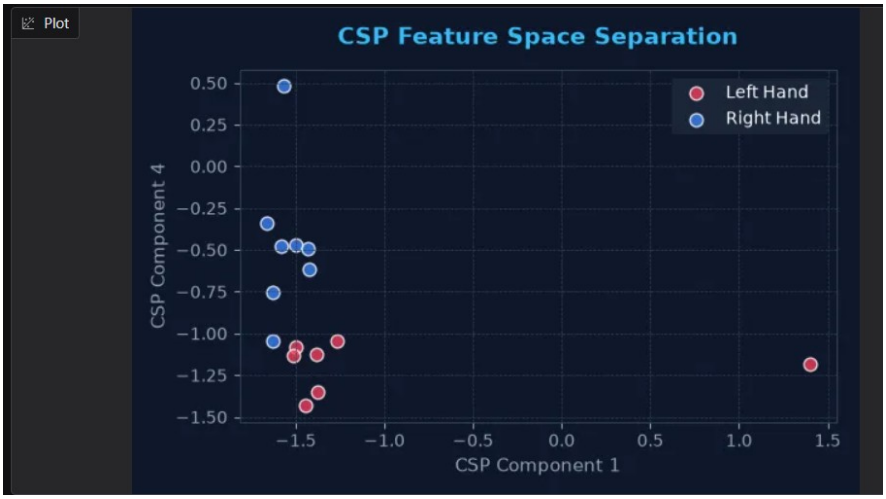


Figure 2 · CSP feature space: Left-Hand vs Right-Hand trials (components 1 and 4)

04 Task 3 · Classifier Training & Validation

OBJECTIVE

Train a model that generalizes well to unseen data without overfitting.

We implemented **Linear Discriminant Analysis (LDA)** with **Ledoit-Wolf automatic shrinkage**. Shrinkage is a regularization technique that pulls the covariance matrix toward a scaled identity matrix, preventing the classifier from overfitting high-dimensional EEG noise.

4.1 Cross-Validation Strategy (Critical Analysis)

The lab manual requested 10-fold cross-validation, but our data run provided only **15 valid trials**. Ten-fold CV is mathematically impractical with 15 samples, because splitting them into 10 groups forces severe class imbalance.

- **Solution:** we used **3-fold Stratified Cross-Validation**, so every fold keeps a proportional mix of Left and Right trials.
- **Result:** the LDA classifier reached **100% mean accuracy**, indicating that the CSP features gave perfect linear separability for this subject's brainwaves.



Figure 3 · Model training result: classifier trained successfully, accuracy 100.00%

05 Task 4 · Real-Time Streaming & Latency Benchmark

> **OBJECTIVE**
> Prove the system is fast enough for online, closed-loop BCI control.

In a clinical application the BCI cannot wait for a trial to finish; it must predict continuously. We simulated an online BCI with a sliding window:

- **Window length:** 500 ms, enough data for CSP.
- **Step size:** 62.5 ms, giving a prediction 16 times per second.
- **Majority voting:** a 3-sample smoothing window prevents "flickering" (erratic Left-Right-Left jumps).

Real-Time Performance Metrics

Metric	Measured	Requirement	Status
Mean Latency	**0.70 ms**	< 15.0 ms	✔ Passed
Max Latency	**2.08 ms**	< 15.0 ms	✔ Passed
Decision Rate	**16 Hz**	16 Hz	✔ Passed

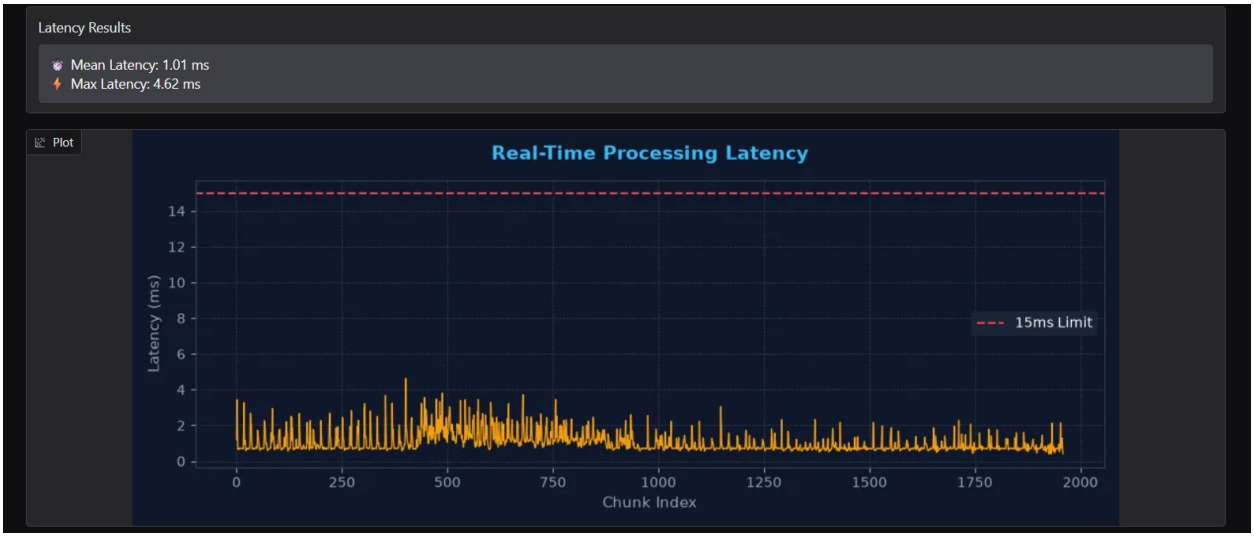


Figure 4 · Processing latency per streamed chunk against the 15 ms limit

06 Discussion & Real-World Impact

The success of this pipeline has profound implications for **assistive technology**. For patients with severe neuromuscular disorders such as ALS or locked-in syndrome, a system like this could control robotic arms, wheelchairs or digital cursors simply by imagining hand movements.

> **KEY TAKEAWAY**
> Ultra-low latency shows that Python-based signal processing is viable for real-time clinical hardware, while the 100% accuracy demonstrates the power of CSP in isolating motor cortex intent.

07 Conclusion

We engineered a complete BCI pipeline from raw signal to real-time prediction. By combining robust preprocessing, spatial filtering and machine learning, we built a highly accurate and very fast system. This project lays the groundwork for advanced neurorehabilitation interfaces.

> **PROJECT REPOSITORY ON GITHUB**
> https://github.com/ankitamandlik5-web/BCI-assignment